SHUBHAM KISHOR KALE

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AI Engineer building production LLM systems, agentic RAG with citation checks and evals, and the data platforms they run on.

WORK EXPERIENCE

Quantegy Analytics | New York, USA **May 2025 - Present**
Analytics / AI Engineer

- Shipped an LLM-powered navigation assistant into a production analytics platform, now answering 150+ questions a day; routed 46 of 50 randomly sampled questions to the right dashboard page.
- Automated intake of 100+ client requests a month with an idempotent Dagster pipeline: an LLM extracts each email into Pydantic-validated JSON; 29 of 30 randomly sampled tickets passed human review unedited.
- Ranked 420 stores for endcap placement with a stacked tree and linear-regression ensemble; the 21 top-ranked stores given endcaps posted +$118K in week-1 sales on a $984K base (+12%) vs +0.25% in 14 controls.
- Built the Snowflake data platforms for 3 clients on Dagster and Airflow: 4M+ healthcare records from Redshift, 120+ EHR tables for a correctional healthcare client, ~5x faster ingestion for an insurance client.

Mettler-Toledo | Mumbai, India **Mar 2024 - Jun 2024**
AI Engineer Intern

- Designed a RAG system with fine-tuned T5/BART models that drafts KPI commentary from 10M+ financial records; 85%+ of drafts rated relevant in a 40-case team review, saving 15 hrs/month of manual write-ups.

Marcellus Investment Managers | Mumbai, India **Mar 2023 - Nov 2023**
Quantitative Analyst Intern

- Developed a regression-based impact-cost model (R-squared 0.82) for trade-execution optimization and a forensic-screening ML ensemble (84% accuracy, 87% specificity) adopted into fraud-risk screening.

PROJECTS

AI Financial Research Agent · [GitHub] · [Live Demo] · *Agentic RAG, MCP, Evals, Observability*

- Architected a LangGraph ReAct agent over SEC 10-K filings with 5 MCP-served tools, including XBRL fact lookup, and per-thread conversation memory (11 of 11 follow-ups in a small probe, 3 of 11 without).
- Validated each served answer with a fail-closed contract, no model in the loop: figure-to-source matching and citation validation against the retrieved evidence, passed by all 71 benchmark answers.
- Lifted retrieval hit@5 from 0.51 to 0.66 on the same benchmark with cross-encoder reranking; XBRL fact tools raised figure accuracy from 73% to 100% on its 45 figure items and RAGAS faithfulness from 0.83 to 0.94.
- Engineered the full stack: FastAPI streaming cited answers over Server-Sent Events to a Next.js UI, LangSmith tracing and a per-query meter (p50 2.8 s, p95 6.9 s, about $0.002); 362 tests, including 33 covering failure paths.

AI Auditor · [GitHub] · [Live Demo] · *Human-in-the-Loop, Multimodal Document Extraction, Guardrails*

- Deployed a 5-stage audit review workspace (FastAPI, React, Supabase) where a vision LLM proposes facts and judgments and deterministic Python owns materiality, sampling and approval, so the model never signs off.
- Implemented structured extraction over PDFs, scans and spreadsheets through a strict JSON-schema contract with a vision model, screened by 16 deterministic, policy-cited evidence checks before a fact reaches a workpaper.
- Enforced controls in code: row-level security on 7 Postgres tables, row-locked versioned approvals, a staleness cascade on corrections, and a model adapter with retries and backoff; 62 tests, 9 of them security regressions.

TECHNICAL SKILLS

Languages, Frameworks & Cloud: Python, SQL, TypeScript, PyTorch, scikit-learn, Hugging Face, Claude API, Gemini API, LangChain, LangGraph, LangSmith, FastMCP, ChromaDB, RAGAS, FastAPI, Next.js, Docker, AWS (S3, Redshift), Snowflake, PostgreSQL, Pydantic, pytest, GitHub Actions (CI/CD), Claude Code, Codex
Concepts & Methods: Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Structured Outputs, LLM Evaluation, LLM Observability, LLM Guardrails, Prompt Engineering, Machine Learning

EDUCATION

Stony Brook University, New York, USA | *M.S. in Data Science* | **GPA: 3.78/4** **Aug 2024 - Dec 2025**

NMIMS University, Mumbai, India | *B.Tech. in Data Science* **Jul 2019 - May 2023**